Program : **Diploma in Robotics and Automation**	
Course Code : **2371**	Course Title: **Foundations of Robotics**
Semester : **2**	Credits: **3**
Course Category: **Program Core**	
Periods per week: **3 (L:3 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To explore the broad scope of robotic applications
- To introduce the functional elements of Robotics
- To understand the basic concepts associated with the design, functioning, applications and social aspects of robots
- To study about the electrical drive systems and sensors used in robotics for various applications
- To develop basic robot construction skills.

Course Prerequisites:

Topic	Course code	Course Name	Semester
Basic principles and theorems of Engineering Physics and Chemistry		Physics and Chemistry	1 & 2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand the broad concepts of robotics and its applications	13	Understanding
CO2	Develop basic kind of autonomous robots	15	Applying
CO3	Select and use sensors in various robotic application contexts	15	Applying
CO4	Select and use actuators in various robotic application contexts	15	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	`	PO7
CO1	3						
CO2		3					
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand the broad concepts of robotics and its applications.**		
M1.01	Understand The fundamentals of Robots Types and Need	1	Understanding
M1.02	Understand the characteristics of Robots	3	Understanding
M1.03	Overview of various Applications of Robots	3	Understanding
M1.04	Industrial Application	4	Understanding
M1.05	Different parts and functions of Robots	2	Understanding
Contents: **Fundamentals of Robots**: Definitions- Robots, Robotics; Need of Robots, Types of Robots, Robot Characteristics, Advantages and Disadvantages of Robots, Robot Applications- Medical, Mining, Space, Defense, Security, Domestic, Entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining. Systems overview of a robot, Robot Parts and their functions			
CO2	**Develop basic kind of autonomous robots.**		
M2.01	Understanding the Basic Principle of a Line Follower	2	Understanding
M2.02	Assembling sensors	2	Applying
M2.03	Assembling the Line Follower	2	Applying
M2.04	Understand Interfacing the Microcontroller with the Motor Driver & the Sensor IC	3	Understanding

M2.05	Understand the Basic Principle of Moving	3	Understanding
M2.06	Assembling the Obstacle Avoider	3	Applying
	Series Test – I	**1**	

Contents:
Assembling the Line Follower - Basic Principle of Color Sensing - Basic Principle of a Line Follower - Assembling of the Line Follower Robot - Assembling of the Sensors - Assembling the Whole System to the Motor Driver - Interfacing the Microcontroller with the Motor Driver & the Sensor IC. **Assembling the Obstacle Avoider** - Basic Principle of Moving - Basic Principle of a Obstacle Avoider - Assembling of the Obstacle Avoider Robot - Assembling of the Sensors - Assembling the Whole System to the Motor Driver - Interfacing the Microcontroller with the Motor Driver & the Sensor IC

CO3	**Select and use sensors in various robotic application contexts.**		
M3.01	Understand the basics of sensors-basic idea	1	Understanding
M3.02	The working of different types of Internal sensors -position Sensors and velocity sensors and their selection	3	Applying
M3.03	The working of different types of external sensors -proximity sensors, force and torques sensors and their selection	3	Applying
M3.04	Different sensing variables and Different Sensors for Robotics	4	Understanding
M3.05	Basics of variable recognition systems	4	Applying

Contents:
Sensors In Robotics - Internal Sensors - Position sensors - optical, non-optical, Velocity sensors, Accelerometers – External Sensors - Proximity Sensors - Contact, non-contact, Range Sensing, touch and Slip Sensors, Force and Torque Sensors Miscellaneous Sensors In Robotics - Different sensing variables - Smell, Heat or Temperature, Humidity, Light, Speech or Voice recognition Systems.

CO4	**Select and use actuators in various robotic application contexts**		
M4.01	Understand the building block of a robot	2	Understanding
M4.02	The application of actuators in building block of a robot and different types of DC motors used as actuators	3	Understanding
M4.03	Speed and direction control of drives and selection of actuators	3	Applying
M4.04	Nontraditional sensors and actuator	2	Applying

M4.05	Optical, inertial, thermal, chemical, biosensor and other common sensors	3	Applying
M4.06	Selection of sensors and actuators for self driving car	2	Applying
	Series Test – II	1	

Contents:
Building Blocks of a robot - Types of electric motors - DC, Servo, Stepper; specification, drives for motors - speed & direction control and circuitry, Selection criterion for actuators, direct drives, non-traditional actuators; Sensors for localization, navigation, obstacle avoidance and path planning in known and unknown environments – optical, inertial, thermal, chemical, biosensor, other common sensors; Case study on choice of sensors and actuators for maze solving robot and self driving cars

Text / Reference

T/R	Book Title/Author
T1	Koren Yoram, Robotics for Engineers McGraw - Hill Education, New Delhi, 1 st Edition
R1	Hedge, G S, Textbook On Industrial Robotics Laxmi Publications, New Delhi, ,1st Edition
R2	Groover Mikell P. Industrial Robotics: Technology, Programming and Applications, McGraw - Hill Education, New Delhi 2 nd Edition
R3	K. K.Appu Kuttan, Robotics, I K International, 2007
R4	Fu K. S., Gonzalez R C., Lee C S G, Robotics. McGraw - Hill Education, New Delhi Pvt. Ltd
R5	Sawney A K and Puneet Sawney, “A Course in Mechanical Measurements and Instrumentation and Control”, 12th edition, Dhanpat Rai & Co, New Delhi, 2013.

Online Resources

Sl.No	Website Link
1	https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-to-robotics-fall-2005/index.html
2	https://nptel.ac.in/courses/112/101/112101098/
3	https://nptel.ac.in/courses/112/105/112105249/
4	https://create.arduino.cc/projecthub/robocircuits/line-follower-robot-arduino-299bae
5	https://create.arduino.cc/projecthub/Alfa0420/obstacle-avoiding-robot-using-arduino-and-ultrasonic-sensor-c179bb